

LABORATORY0 General Lab Guide

Welcome to EE111L!

“Fundamentals of Electric Circuits” may be your perfect guide to the marvelous world of electronics. So, let’s do something interesting in the coming labs! The lab0 is extremely important, please read patiently and **carefully**. In this general guide, you will find information about how to go through labs, as well as logistics such as grading policies.

Each lab assignment will usually consist of two components: pre-lab and lab. In the pre-lab, we will mainly cover the basic theory that will be presented in the lab. You must have learned some knowledge in the class before the labs. Please make sure that you figure out the theory in your textbook. During your time in the laboratory, you will actually build the circuits and explore the practical and non-idealized implementation of the circuit material presented. Please note that, **the pre-lab report and lab report is completed and submitted independently.**

Pre-Lab

Good laboratory practice begins with pre-lab preparation. The Pre-lab Preparation section contains several tasks that must be performed BEFORE arriving at the lab. It is essential to read the lab manual, perform the pre-lab assignments, and carefully think through all the steps to be performed and the measurements to be made. The pre-lab typically requires calculation of specific component values, prediction of the behavior to be measured in the lab, or preparation using computer simulation. If you do not complete the pre-lab, there is a good chance that you will NOT be able to complete the lab in the allotted lab section. Carefully read lab experiment handouts prior to start of the lab. Any ambiguity encountered must be brought to the attention of the instructor and TA immediately. **The pre-lab report should be completed individually.**

Lab

Students will work in teams of 2 in the laboratory. These teams should remain the same throughout the semester. We would like to remind you that your behavior in the lab is also a part of every lab score. You must act responsibly at all times for your own safety and the safety of others. If you have any misconducts, we reserve the right to take some points off. The detail is shown in Code of conduct for the laboratories.

Code of conduct for the laboratories

You must act responsibly at all times for your own safety and the safety of others.

1. Be certain that your hands are dry and that you are not standing on wet floor.
2. Eating, Drinking or Smoking is not permitted in the Lab.
3. Do not wear sandals/slippers in Lab.
4. ATTENDANCE: Attendance is mandatory and any absence must be for a valid excuse and must be documented. No attendance, no scores.
5. The lab timetable must be strictly followed.
6. Be PUNCTUAL for your lab time.
7. Experiment must be completed within the given time.
8. Equipment or furniture in the Lab should not be relocated without permission.
9. Workspace must be kept clean and tidy at all time.

10. All equipment, apparatus, tools and components must be RETURNED to their original place after use.
11. All items, which belong to the Lab such as equipment, tools and furniture, are strictly PROHIBITED to be taken away from the Lab.
12. When equipment has malfunctioned or not working properly, such as smell or burning parts, you should turn off the power and unplug the equipment if it is safe to do so. Immediately exit and report to the technical staff-in-charge and never fix the problems yourself. Report all injuries and accidents to the staff-in-charge immediately.
13. In case of fire incident or other emergency situations, stop working and switch off equipment/devices if it is safe to do so. Follow the EXIT sign to find the nearest EXIT door to evacuate. Inform the staff-in-charge or security immediately.
14. If the Lab has any facility related issues, such as water leak, soiled floor, unusual smell, or door lock not working, please inform teacher immediately.
15. Please check the blackboard and WeChat group regularly for updates

Before leaving the lab

1. After completing the experiment show your readings to the teacher or TA, and switch off the power to your circuit after getting approval from the teacher or TA.
2. Place the stools under the lab bench.
3. Turn off the power to all instruments.
4. Turn off the main power switch to the lab bench.
5. Tidy up the bench and check out then.

Lab Report Submissions

Each laboratory consists of a Prelab and a Lab Report. Lab reports are an essential part of all laboratory courses and usually a significant part of your grade. For each lab, we will provide you with one of these lab documents that will guide you through the lab. At the top of each lab document there will be a box with fields for you and your partner's names and student ID numbers. Typically, the lab for week N will be due the day before the week N+1 unless otherwise specified.

When you submit your lab documents, make sure to attach graphs, circuit diagrams, etc., the more detail the better. It is hard to give partial credit for a lab report that refers to a non-existent circuit diagram. It's simply a reminder that many students are prone to missing the units of their measurements. In these labs, when you are prompted for a numerical answer, make sure to put the correct units.

Policy on Reports

1. Please refer to the course syllabus for the academic integrity statement.
2. Although students are encouraged to collaborate during lab, each must individually prepare a report and submit by the due date even if you are working with a partner.
3. Reports must be prepared in English; you need to show your work in your submissions to receive any credit.
4. Late work should also be sent to the TA (chengle2024@shanghaitech.edu.cn) and cc instructor (lijuan1@shanghaitech.edu.cn) by email. Unless otherwise stated, late work is accepted with a 10-point grade deduction for the first late day and a 10-point grade deduction for each additional late

day, which means that if you submit your work a week or more days later, you will receive zero points.

Policy on Physical Submission of Reports

Please observe the following guidelines:

1. All paper reports must be submitted to the collection box located in SIST 1D105.
2. Ensure your name, student ID number, and group number (Table No.) are clearly indicated on the first page.
3. Depositing your assignment into the box constitutes official submission. Materials cannot be returned or retrieved.

Policy on Electronic Submission of Reports

1. Please submit **electronic copies** of your work in PDF format to Blackboard.
2. For each assignment, you have three attempts to submit your work before ddl. We will use the report you last submitted as the final submission.
3. The file naming format is to add your name after the file name, taking the Prelab/Labx report as an example:

Table No._Prelab/Labx_202xxxxxxx_ Your Chinese Name.

Grading

This course is graded Pass or No Pass. To pass the course, students must meet both the attendance requirements through ongoing active participation and the 60% or higher grading requirement.

The general distribution of grading is shown below. All assigned work must be submitted in order to receive a grade. Please note That, No Experiment, No Score, even if you hand in the Lab Report.

ACTIVITIES	PERCENTAGES
Prelab	3%*7
Lab (Experimental operation & Lab Reports, Lab Policy)	9%*7
Project	16%

Asking for Help

Instructor: Li Juan (lijuan1@shanghaitech.edu.cn)

TA: Cheng Le (chengle2024@shanghaitech.edu.cn)